

Sparse Congruence-Preserving Perturbations of Dyadic Series

Countermodels and integral-shift criteria for Erdős Problem #251

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ABSTRACT

Let $A = \sum_{n \geq 0} a_n 2^{-(n+1)} < \infty$ with nonnegative integer coefficients. For every prescribed cutoff and every $f(n) \rightarrow \infty$, we construct a fixed support of upper Banach density zero and a fixed nondegenerate interval above A . Every point of that interval is realised by a nonnegative integer correction supported on this set and eventually bounded by f , while every fixed modulus eventually divides both the corrections and their cumulative sums. For $f(n) = (\log(n+3))^\varepsilon$, the empirical laws of literal gap blocks of length $o(\log \log X)$ are preserved in total variation. The construction combines finite-choice interval covering with congruence buffers. Applied to prime gaps, it shows that these congruence and block-law data, even together with fixed-block polynomial nonconcentration and the cumulative prime-growth scale, do not distinguish rational from irrational dyadic sums. It does not preserve primality. We also give exact integral-shift criteria for the actual prime-gap tails and isolate the joint small-shift recurrence that would suffice to prove irrationality. That prime-specific input is not established here.

1 Introduction

Sparse corrections and local information. Our main result concerns general summable nonnegative integer words, not only prime gaps. The support and target interval are chosen before the target value. A congruence buffer repairs the cumulative residue; a pair with fixed unweighted total then changes the dyadic value without undoing that repair. Sparsity ensures that almost every sufficiently short block meets no correction.

Density and block conventions. Here $\mathbb{N} = \{0, 1, \dots\}$, all intervals of indices contain integers, and $\log$ is natural unless a base is displayed. Upper Banach density zero means

$$\lim_{H \rightarrow \infty} \sup_{u \in \mathbb{N}} \frac{|S \cap [u, u+H)|}{H} = 0.$$

For integers $X \geq 1$ and $m \geq 1$, let $\mu_{a,X,m}$ be the empirical probability measure giving mass $1/X$ to each literal block $(a_n, \dots, a_{n+m-1})$, $X \leq n < 2X$, with multiplicities. We

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

use $d_{\text{TV}}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|$; consequently tests bounded in absolute value by B_0 differ in mean by at most $2B_0 d_{\text{TV}}(\mu, \nu)$. No rescaling of the coefficients is implicit.

Proposition 1.1 (sparse congruence-preserving rationalisation). *Let $a : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $A = \sum_{n \geq 0} a_n 2^{-(n+1)} < \infty$, let $K \in \mathbb{N}$, and let $f : \mathbb{N} \rightarrow \mathbb{R}$ tend to $+\infty$. There exist a set $S \subseteq [K, \infty)$ of upper Banach density zero and a nondegenerate interval $I \subset (A, \infty)$ such that, for every $r \in I$, there is an integer correction $e : \mathbb{N} \rightarrow \mathbb{N}$ satisfying*

$$\text{supp } e \subseteq S, \quad e_n \leq f(n) \text{ eventually}, \quad \sum_{n \geq 0} (a_n + e_n) 2^{-(n+1)} = r.$$

For each integer $q \geq 1$ there is a cutoff N_q , independent of r , such that $q \mid e_n$ and $q \mid \sum_{i < n} e_i$ for all $n \geq N_q$. For every $\varepsilon > 0$, the choice $f(n) = (\log(n+3))^\varepsilon$ can be made with

$$|S \cap [X, 2X]| = O_\varepsilon \left(\frac{X}{\log \log X} \right), \quad d_{\text{TV}}(\mu_{a,X,m}, \mu_{a+e,X,m}) \leq \frac{m|S \cap [X, 2X+m]|}{X}.$$

In particular the distance tends to zero for every integer-valued $m = m(X) \geq 1$ with $m(X) = o(\log \log X)$, uniformly over target values.

Construction. Use disjoint triples of coordinates $n_j - 1, n_j, n_j + 1$, with spacings $s_j = n_j - n_{j-1} \geq 4$, and moduli M_j forming a divisibility chain. Put $D_j = 2^{s_j} - 1$. The integer C_j is the total correction in the completed triples before stage j . Starting from $C_0 = 0$, define

$$c_j = (-C_j) \bmod M_j \quad (0 \leq c_j < M_j), \quad C_{j+1} = C_j + c_j + M_j D_j.$$

For a free integer digit $d_j \in \{0, \dots, D_j\}$, assign all three corrections:

$$e_{n_j-1} = c_j, \quad e_{n_j} = M_j d_j, \quad e_{n_j+1} = M_j(D_j - d_j),$$

and set $e_n = 0$ elsewhere. The complete triple has total $c_j + M_j D_j$, independent of d_j , so every buffer is fixed before the free digits are chosen. Immediately after the buffer, the cumulative correction is divisible by M_j ; both pair entries preserve it. Later buffers preserve every previously imposed divisor. Thus each fixed modulus eventually divides both individual and cumulative corrections if it eventually divides M_j . More precisely, if $q \mid M_j$, every index $n \geq n_j$ has both divisibilities. The first buffer at $n_j - 1$ need not itself be divisible by q ; it is deliberately before this cutoff. The cutoff depends only on the schedule, not on the chosen free digits.

Here is an explicit schedule for an arbitrary $f \rightarrow \infty$. Set

$$h(n) = \inf_{m \geq n} \min(f(m), m),$$

choose $n_{-1} \geq K$ so large that $h(n_{-1}) \geq 32$, and recursively let

$$k_j = \max\{k \geq 2 : k! 2^{k+2} \leq h(n_{j-1})\}, \quad M_j = k_j!, \quad s_j = k_j + 2, \quad n_j = n_{j-1} + s_j.$$

The function h is nondecreasing, tends to infinity, and satisfies $h(n) \leq n$. Hence k_j is nondecreasing and tends to infinity. Each triple entry is at most $M_j 2^{s_j} \leq h(n_{j-1})$, which is at most $f(n)$ at every coordinate of that triple and at most n_{j-1} . This proves the size

bound and summability of the correction. To see uniform sparsity, fix R . Beyond finitely many triples, centres are separated by at least R . An interval of length H therefore meets at most $3(H/R + 2)$ supported coordinates, plus a fixed finite number. Divide by H , take the supremum over translates, let $H \rightarrow \infty$, and then let $R \rightarrow \infty$.

The fixed total of the pair is the point of using two free coordinates: for example, $(6d, 6(15 - d))$ always has ordinary sum 90, but advancing d by one changes its dyadic contribution by $6 \cdot 2^{-n_j-2}$. In general put $w_j = M_j 2^{-n_j-2}$. The weighted correction separates as

$$\sum_n \frac{e_n}{2^{n+1}} = \beta + \sum_j d_j w_j, \quad \beta = \sum_j (c_j 2^{-n_j} + D_j w_j).$$

The buffer contribution is included in the fixed, positive number β . For $i > j$, monotonicity of the moduli gives

$$D_i w_i \geq M_j (2^{-n_{i-1}-2} - 2^{-n_i-2}), \quad \sum_{i>j} D_i w_i \geq w_j.$$

For each j let $X_j = \{0, w_j, \dots, D_j w_j\}$. Its maximal successive gap is w_j , while its diameter is $D_j w_j$. Thus the last displayed inequality is exactly the finite-choice covering condition

$$\text{gap } X_j = w_j \leq \sum_{i>j} \text{diam } X_i.$$

Here $\text{gap } X$ denotes the largest gap between consecutive points of a finite ordered set X . Crmarić and Kovač's covering lemma [20, Lemma 4] therefore gives $\sum_j X_j = [0, F]$, where $F = \sum_j D_j w_j > 0$. Equivalently, choose digits successively so that each remainder lies within the remaining capacity, which tends to zero. This is a finite-menu form of Fridy's generalised-base argument [6, Lemma, p. 194]; unlike Fridy's statement, the finite-menu formulation needs no monotonicity of (w_j) . Reciprocal menus appear in Kovač–Tao [7, Lemma 5.1]. Take $I = (A + \beta, A + \beta + F)$.

For the polylogarithmic bound, instead set, after a sufficiently large initial cutoff,

$$s_j = \left\lfloor \frac{\varepsilon}{2} \log_2 \log(n_{j-1} + 3) \right\rfloor, \quad k_j = \max\{k \geq 2 : k! \leq (\log(n_{j-1} + 3))^{\varepsilon/4}\},$$

with $M_j = k_j!$, $n_j = n_{j-1} + s_j$ and the same D_j and buffers. This schedule works for every $\varepsilon > 0$; no prime-growth assertion is used here. Then $s_j \geq 4$ and $M_j 2^{s_j} \leq (\log(n_{j-1} + 3))^{3\varepsilon/4}$. The preceding congruence, convergence and interval arguments still apply. Spacing comparable to $\log \log n_j$ gives $|S \cap [X, 2X]| = O_\varepsilon(X / \log \log X)$. Couple the original and corrected length- m blocks at the same starting index in $[X, 2X]$. At most $m|S \cap [X, 2X + m]|$ starts meet a correction, so their empirical distributions differ in total variation by $O_\varepsilon(m / \log \log X)$ for $m = o(\log \log X)$. The companion record expands this three-coordinate construction and its all-index congruence check. The pinned formal development proves the same existential conclusion by a different, one-site residue-feedback construction. Its theorem statements, rather than identity of witnesses, are the formal correspondence; the verification boundary is recorded below. $\square$

For a fixed bound B_0 , this coupling also applies to tests $\Phi_X(n, a_n, \dots, a_{n+m-1})$ depending on the starting index: their average difference is at most $2B_0 m |S \cap [X, 2X + m]| / X$.

This is an absolute error estimate. For an event of probability $p_X \rightarrow 0$, relative preservation would require this error to be $o(p_X)$; it does not follow from an error of $o(1)$. A cofinal but density-zero witness set can be lost altogether.

Corollary 1.2 (one rational countermodel with joint constraints). *Let $p_0 = 2, p_1 = 3, \dots$ be the primes and $g_n = p_{n+1} - p_n$. Given $K \in \mathbb{N}$ and $0 < \varepsilon \leq 1$, there is $b : \mathbb{N} \rightarrow \mathbb{N}$ with rational dyadic sum such that $b_n = g_n$ for $n < K$, $b_n \geq g_n$, and $b_n - g_n \leq (\log(n+3))^\varepsilon$ eventually. For every fixed positive modulus, both the coefficients and the cumulative positions eventually retain their corresponding residues. The literal block laws agree asymptotically for lengths $o(\log \log X)$, and for every fixed nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$,*

$$|\{n < N : F(b_n, \dots, b_{n+k}) = 0\}| = o(N).$$

Moreover, with $P_n = 2 + \sum_{i < n} b_i$,

$$0 \leq P_n - p_n = O_\varepsilon\left(\frac{n(\log(n+3))^\varepsilon}{\log \log n}\right), \quad P_n \sim n \log n.$$

These positions are not asserted to be prime.

Proof. Choose a rational target in Proposition 1.1's interval and put $b = g + e$. Fixed-block nonconcentration for g is Schlage-Puchta's Lemma 4 [26]. A length- $(k+1)$ block changes only at a start in $\bigcup_{i=0}^k (S - i)$, a density-zero set. Thus the same zero-density conclusion holds for each fixed polynomial at b , without a bound on its values or on the values of e . Summing the dyadic support bound gives $|S \cap [0, n]| = O_\varepsilon(n / \log \log n)$. Multiply by the eventual pointwise bound on e to obtain the displayed cumulative estimate. The prime number theorem gives $p_n \sim n \log n$ [13, Chapter 6]; the error is $o(n \log n)$ even at $\varepsilon = 1$. The congruences and growing-block conclusion are those of the proposition. $\square$

This corollary is an ordinary deduction combining the sparse construction with an external nonconcentration theorem. It is not a claim that the combined statement, or Schlage-Puchta's analytic input, was newly checked in Lean. The bounded, single-modulus alternative and its translation argument remain in the companion record.

Relation to prior work. Erdős's fixed-denominator question appears in [1, p. 94], [2, p. 62] and [4, p. 103]. Neither his factorial-denominator theorem nor the variable-denominator counterexample in [18] decides it. The bounded prime-factor comparison digits of Erdős–Pomerance [3, §7] are a different dyadic setting.

The closest mechanism for Proposition 1.1 is the finite-choice covering lemma of Crmarić–Kovač [20, Lemma 4], which they use for product-denominator series. Fridy [6] and Kovač–Tao [7, Lemma 5.1] provide generalised-base and reciprocal-menu antecedents. Van Doorn–Kovač combine interval filling with a chain absorbing every integer divisor [14, Lemma 7, Proposition 8]. Our factorial moduli use that absorption property. Van Doorn's reciprocal-preserving exchange [22, proof of Theorem 3] is a useful comparison: our pair instead preserves an ordinary sum while varying its dyadic weight. These are method comparisons, not claims that the present perturbation theorem is a special case of those arithmetic statements, or that interval filling is new.

Finite binary menus can have more complicated achievement sets. Central blocks of subsums can give intervals even when elementary term-versus-tail tests alternate [25, Theorem 2]; Cantorvals may occur when the overlap inequality $a_n \leq \sum_{i>n} a_i$ holds only on a density-zero set of indices [24, Theorem]. See [23, §1] for background, and [29, Theorem 14 of the consulted preprint] for Nitecki’s exposition of the Guthrie–Nymann classification. The repaired constructions in [30, Theorem 2] show that even a prescribed pattern of strict term-versus-tail inequalities need not determine interior. Nowakowski’s different sufficient condition requires an infinite positivity procedure, not just mixed inequalities [31, Definition 2, Theorem 3.1]. Our proof avoids that classification issue by verifying a covering inequality for every finite menu.

There is a separate boundary concerning prime statistics. Schlage-Puchta’s fixed-block polynomial nonconcentration lemma [26, Lemma 4] is not a growing-block distribution theorem. Adamczewski–Drmota–Müllner give effective logarithmic densities for fixed automatic sequences along primes and a criterion for natural densities [21, Theorems 1.2, 1.4]; this does not describe the unbounded consecutive-gap word or a family of automata growing with X . In particular, logarithmic densities in their general theorem need not be rational.

Under a uniform Hardy–Littlewood hypothesis of the type stated by Kuperberg [8, Conjecture 1.3], Land’s draft proves a conditional irrationality implication [9, Theorem 2] and Ringer’s draft a conditional normality implication [19]. The authors supply formalisation material; no independent rebuild of those external projects is claimed here. Ringer also separates weaker averaged comparison inputs from the stronger uniform conjecture. None of these conditional hypotheses is assumed in this note, and no unconditional irrationality claim is made.

Reading the two papers. This note gives the sparse construction, its joint consequence and the exact prime-tail criterion. The companion *Prime-Gap Dyadic Series: Perturbations, Exact Criteria and Certificates* supplies the extended algebra, complete countermodels, source concordance and certificate code. The manuscripts use stable labels rather than section numbers as the cross-reference contract.

2 The prime series and its actual tails

Retain the zero-based primes and gaps of Corollary 1.2, and put

$$\Pi = \sum_{n \geq 0} p_n 2^{-(n+1)}, \quad G = \sum_{n \geq 0} g_n 2^{-(n+1)}.$$

Convergence needs no prime number theorem. The central-binomial argument of Erdős [27, pp. 194–196] gives the elementary bound $p_n \leq 1250(n+1)^4$: for $m \geq 4$, $4^m < m \binom{2m}{m} \leq m(2m)^{\pi(2m)}$. The first inequality follows by induction from $m = 4$; each prime-power contribution to the binomial coefficient is at most $2m$, giving the second. Taking $m = (n+5)^4$, a supposition $\pi(2m) \leq n$ would give

$$m(2m)^n = 2^n(n+5)^{4(n+1)} \leq 4^m,$$

since $x \leq 2^x$ for $x = n+5$ and $n+4(n+1)x = 4x^2 - 15x - 5 \leq 2x^4$. This is a contradiction. Hence $p_n \leq 2(n+5)^4 \leq 1250(n+1)^4$; both series converge absolutely.

Theorem 2.1 (prime-to-gap identity). *The actual prime and gap series satisfy $\Pi = 2 + G$. Their complete tails*

$$T_N = \sum_{j \geq 1} g_{N+j} 2^{-j}$$

satisfy $T_0 = 2G - 1$ and $T_{N+1} = 2T_N - g_{N+1}$. Thus Π , G and T_0 have the same rationality status.

Proof. Finite summation by parts retains the endpoint:

$$\sum_{i=0}^n \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}}.$$

The endpoint tends to zero by the polynomial bound. Splitting off the first term of the absolutely convergent tail gives its recurrence, and $g_0 = 1$ gives $T_0 = 2G - 1$. $\square$

This is the reduction noted by Tao in the problem's forum discussion [17, 7 October 2025], with convergence and the endpoint made explicit; summation by parts itself is not a new method.

Lemma 2.2 (the boundary condition identifying a true tail). *Let $\sum_{j \geq 1} |a_j| 2^{-j} < \infty$ and $U_{N+1} = 2U_N - a_{N+1}$. Then $U_N = \sum_{j \geq 1} a_{N+j} 2^{-j}$ for every N if and only if $2^{-N} U_N \rightarrow 0$.*

Proof. The recurrence gives $2^{-N} U_N = U_0 - \sum_{j=1}^N a_j 2^{-j}$. Its limit vanishes exactly when U_0 is the full series. Without this boundary condition an arbitrary homogeneous term $C2^N$ remains. $\square$

3 Integral shifts: an exact algebraic classification

A real integer-digit recurrence is a sequence $U : \mathbb{N} \rightarrow \mathbb{R}$ satisfying $U_{N+1} = 2U_N - a_{N+1}$, with integers a_n . Write $D_h(N) = U_{N+h} - U_N$. This general recurrence is distinct from the actual prime-gap tail T until the latter is explicitly substituted.

Theorem 3.1 (exact rationality classification). *For a real integer-digit recurrence, the following are equivalent: $U_0 \in \mathbb{Q}$; $D_h(N) \in \mathbb{Z}$ for some $h \geq 1, N \geq 0$; and, for some fixed $h \geq 1$, $D_h(N) \in \mathbb{Z}$ at every sufficiently large N . More precisely, if $U_0 = u/(2^s d)$ is in lowest terms, with d odd, then*

$$\text{den}(U_N) = 2^{\max(s-N, 0)} d, \quad D_h(N) \in \mathbb{Z} \iff N \geq s \text{ and } d \mid 2^h - 1 \quad (h \geq 1).$$

Consequently U_0 is irrational exactly when every positive shift is nonintegral at every index, equivalently when for each fixed $h \geq 1$ there are arbitrarily late nonintegral shifts.

Proof. Iteration gives integers $B_N, C_{N,h}$ with

$$U_N = 2^N U_0 - B_N, \quad D_h(N) = 2^N (2^h - 1) U_0 - C_{N,h}.$$

The second coefficient is nonzero for $h \geq 1$, so one integral shift forces U_0 rational. For rational U_0 , cancellation in lowest terms gives the displayed denominator and divisibility formulae, since $2^h - 1$ is odd. Euler's congruence supplies $d \mid 2^{\varphi(d)} - 1$; for $d = 1$ take $h = 1$. This gives an integral fixed shift for every $N \geq s$. Negating the pointwise and eventual statements gives the last assertion. $\square$

For instance $U_0 = 1/12$ and all digits zero gives $U_2 = 1/3$. The first admissible shift length is 2, and it is integral from index 2 onwards. This illustrates why a test at one preassigned shift length cannot exclude every rational denominator.

Applied to Theorem 2.1, this is an equivalence for the actual prime series, not an irrationality proof. The lcm diagonal, totient witness, forward propagation, free-pair formulation and affine-lattice reformulations are developed in the companion record. Each uses the same integer-orbit calculation; none adds prime-distribution information.

4 A local certificate and the missing prime input

For the actual tail put $D_N = T_{N+h} - T_N$ and $\delta_N = g_{N+h+1} - g_{N+1}$. Then $D_{N+1} = 2D_N - \delta_N$. The gaps in δ_N are even for $h \geq 1, N \geq 0$.

Proposition 4.1 (signed two-window normal form). *Let $D, D' \in \mathbb{R}$, $\delta \in 2\mathbb{Z}$ and $D' = 2D - \delta$. The conditions $|D| < 1$, $|D'| < 1$ and $\delta \neq 0$ hold exactly when, for some $s \in \{-1, 1\}$,*

$$\delta = 2s, \quad \frac{1}{2} < sD < 1.$$

In that case $sD' \in (-1, 0)$ and both states are nonintegral.

Proof. The two unit windows imply $-3 < \delta < 3$. Nonzero evenness forces $\delta = \pm 2$, and substitution gives the signed half-window. Conversely $sD' = 2sD - 2 \in (-1, 0)$ recovers the second unit window. $\square$

Problem 4.2 (cofinal adjacent small mismatch). For every integer $h \geq 1$ and every cutoff N_0 , exhibit $N \geq N_0$ with

$$|T_{N+h} - T_N| < 1, \quad |T_{N+h+1} - T_{N+1}| < 1, \quad g_{N+h+1} \neq g_{N+1}. \quad (1)$$

This would prove irrationality by Theorem 3.1. For each fixed h , Schlage-Puchta's Lemma 4 [26], applied to $x_h - x_0 - 2$ and $x_h - x_0 + 2$, shows that the eligible $\delta_N = \pm 2$ indices have density zero. A proof therefore needs arbitrarily late witnesses in a sparse set, not a positive-proportion statement. The quantifiers are per fixed h ; constants may depend on h .

4.1 Finite, exact tests

For $h \geq 1, N \geq 0, L \geq 1$, put

$$S_{h,N,L} = \sum_{j=1}^L (g_{N+h+j} - g_{N+j}) 2^{-j}, \quad R_{h,N,L}(M) = \sum_{j>L} (M(N+h+j) + M(N+j)) 2^{-j},$$

where $M(n) \geq g_n$ is a nonnegative summable dyadic majorant. Then $|D_N - S_{h,N,L}| \leq R_{h,N,L}(M)$.

Proposition 4.3 (finite separation criterion). *If for every $h \geq 1$ and every cutoff N_0 there are $N \geq N_0, L \geq 1$ with*

$$\text{dist}(S_{h,N,L}, \mathbb{Z}) > R_{h,N,L}(M), \quad (2)$$

then Π is irrational.

Proof. An integral D_N would contradict its distance bound. The resulting cofinal nonintegrality is the criterion in Theorem 3.1. $\square$

Remark (what arbitrary truncation does not gain). Whenever $|D - S_L| \leq R_L \rightarrow 0$, nonintegrality of D is equivalent to $\text{dist}(S_L, \mathbb{Z}) > R_L$ for some L . Indeed, if $\delta = \text{dist}(D, \mathbb{Z}) > 0$, choose $R_L < \delta/2$ and use the 1-Lipschitz property of distance. Thus an unrestricted choice of depth is an exact reformulation, not an additional analytic estimate. A logarithmic depth requires quantitative separation from the relevant boundaries as well as a small remainder.

There is a one-tail version of the adjacent certificate. If $\delta_N = 2s$, $s \in \{-1, 1\}$, and $|QD_N - A| \leq B$ with integers $Q > 0, B \geq 0$, it suffices to check

$$2sA - Q > 2B, \quad Q - sA > B.$$

These inequalities put the entire enclosure of sD_N inside $(1/2, 1)$. For $h = 1, N = 2$, one can use

$$Q = 2^{40}, \quad A = -662838684750, \quad B = 11764181250, \quad s = -1.$$

The two strict margins are 202637379224 and 424908761776. Here $A/Q = S_{1,2,40}$ and B/Q is justified by the polynomial majorant

$$P(x) = x^4 + 8x^3 + 36x^2 + 104x + 150, \quad R_{h,N,L} \leq \frac{1250}{2^L} (P(N+h+L+2) + P(N+L+2)).$$

The identity $2P(x) = (x+1)^4 + P(x+1)$ telescopes the weighted quartic tail; the bound $g_n \leq p_{n+1} \leq 1250(n+2)^4$ completes its proof. The companion record and supplied integer program give all prime indices, both original enclosures and the new one-tail test. One certified pair says nothing about arbitrarily late pairs or all h .

5 What the countermodels exclude

The joint countermodel of Corollary 1.2 shows that the stated prefix, congruence, growth, nonconcentration and growing-block properties can hold together while the dyadic value is rational. The obstruction is to deductions from those properties alone, not to arguments using exact primality or more accurate rare-event estimates. The ordinary proof can also place its target interval inside $(A, A + \eta)$ for any $\eta > 0$: start beyond K' with $(K' + 1)2^{-K'} < \eta$ and use $e_n \leq n$ from the general schedule.

A simple comparison displays the carry freedom explicitly. Set $c_n = 2(n^2 + 4n + 2)$ and $U_n = 2(n+4)^2$. Then $U_{n+1} = 2U_n - c_{n+1}$, every shift is integral, and

$$\sum_{j \geq 1} c_j 2^{-j} = 32, \quad \sum_{n \geq 0} c_n 2^{-(n+1)} = 18.$$

The digits are positive, even and strictly increasing, and their adjacent differences $4n+10$ are never ± 2 . Thus nonperiodicity of unbounded digits does not imply irrationality of a dyadic sum. This is a synthetic countermodel, not a prime-gap construction; the companion gives the full polynomial and carry statements and their formal locators.

Bounded gaps [15], bounded clusters of each fixed size [12, 28], and extreme large gaps [5] concern different distributional data. In particular Polymath’s $\liminf_n (p_{n+1} - p_n) \leq 246$ [28, Theorem 1.4(i)] improves the earlier bound 600; no exact small-gap bound used here supplies a signed weighted-tail window. With the polynomial majorant, $L = \lceil (4 + \eta) \log_2 (N + h + 2) \rceil$ for fixed h makes the remainder small. It certifies a window only when the finite sum also has a larger margin from the boundary. A fixed positive margin is a convenient stronger sufficient condition, not a consequence of strict membership in an open window.

The remaining task is a quantitative prime-gap estimate producing the required separation or sparse signed-window witnesses. The construction, classification and finite certificates above do not supply it. The companion includes the free-pair alternative, affine-lattice obstructions, the complete countermodels and separate denominator exclusions. A lower bound on the denominator of a hypothetical rational value is neither a proof of irrationality nor an irrationality-exponent estimate.

Attribution and verification

The finite-choice covering step is credited where it is used. The sparse existential results and recurrence criteria have counterparts at the pinned source revision 99f4bf47422a; the supplied index marks the named main endpoints `ci_checked`. The buffered-triple witness printed here differs from the one-site formal construction. The new joint corollary and explanatory deductions are ordinary proofs; Schlage-Puchta’s lemma and the prime number theorem are external inputs. This revision replayed `lake build ErdosProblems Erdos249257` on Lean 4.29.1; no new #251 endpoints were admitted beyond that replay. The companion provides the declaration-level ledger and preserves all statements moved out of this shorter presentation.

Lean 4 [10] and `mathlib` [11] provide the proof-checking environment. A supplied build receipt, an ordinary argument and an executed exact-integer certificate have different evidential roles. The original production provenance macro remains required for publication. The author received no external funding and declares no competing interests. The numbering follows Bloom’s catalogue [16]; private review material and future correspondence imply no specialist endorsement.

6 Pinned Lean sources

The following declarations are this note’s pin-faithful source inventory. Line numbers are those of the commit named by `\commit`.

- [prime gap0 zero](#)
- [prime gap0 one](#)
- [dyadic partial sum q](#)
- [dyadic difference partial sum q](#)
- [dyadic partial sum q eq start add differences](#)
- [prime0 displayed partial sum q eq two mul](#)

- prime0 dyadic summation by parts
- prime dyadic term
- prime gap dyadic term
- prime gap dyadic term eq
- summable prime gap dyadic term of summable prime dyadic term
- tsum prime dyadic term eq two add prime gap
- prime displayed dyadic term
- prime displayed dyadic term eq two mul
- irrational tsum prime dyadic term iff prime gap
- tsum prime displayed dyadic term eq four add two prime gap
- irrational tsum prime displayed dyadic term iff prime gap
- prime0 le polynomial
- summable prime dyadic term
- tsum prime dyadic term eq two add prime gap unconditional
- dyadic tail recurrence
- tail shift
- dyadic tail block
- tail iterate eq pow mul sub block
- tail shift eq scaled sub block
- tail shift succ
- rat integral sub int iff
- tail shift integral iff scaled tail
- tail shift integral iff den dvd mersenne
- rat integral totient multiplier of odd den
- tail shift integral totient of odd den
- tail shift integral succ
- tail shift integral add
- tail den succ
- tail den succ eq of odd
- tail den succ eq half of even
- rat integral scaled of den eq pow two mul odd

- rat integral multiplier tail at two exponent
- rational prime gap tail shift eventually integral of den eq
- rational prime gap tail shift eventually integral
- rational prime gap tail state
- rational prime gap tail state recurrence
- exists rational prime gap tail state representation of not irrational
- not irrational initial iff exists integral positive tail shift
- not irrational initial iff exists eventually integral positive tail shift
- irrational initial iff all positive tail shifts nonintegral
- irrational initial iff cofinal nonintegral tail shifts
- tail shift not both integral of small pair of digit ne
- not eventually integral tail shift of cofinal small mismatch
- prime gap tail shift not eventually integral of cofinal small mismatch
- exists prime gap0 gt
- prime gap0 not eventually periodic
- carry coeff
- carry partial sum eq
- rational prime gap tail has positive shift not eventually small
- prime0 displayed partial sum q
- dyadic partial sum q succ
- dyadic difference partial sum q succ
- prime0 mono step
- prime gap0 cast
- prime gap partial sum q
- carry partial sum
- nat carry coeff
- nat carry coeff cast

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